CCDC and AI/ML-Ready Solar Datasets: Data-Science Analysis

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1. Executive summary

- **The CCDC page** (`https://nature.njit.edu/solardb/ccdc`, opened 2026-10-13; snapshot in `papers/citations/`) is titled **"CCDC: Community Coordinated Data Center"**. It states: "These AI-ready datasets are now available on GitHub and Zenodo for interested researchers to download and use."
  - It currently lists **one dataset: FlareDB**. The card links to `github.com/Reasopprime/njit-flaredb`.
  - Its Citation button points to Zenodo doi:10.5281/zenodo.16790538 (quick-look movies, version 2025-05-31).
  - The dataset paper is Liu et al. 2026, *Scientific Data* (open access, read in full).
  - FlareDB is profiled in §2.1.
- **Also analysed:** the ML-ready datasets distributed with the 19 CCSC tools (more than 12 tables and image sets, inspected directly), the MagNet validation package on SolarDB, and published descriptions of other SolarDB datasets and of the partner dataset SolARED.
- **Main data-science findings** (details in §4–§5):
  1. **Provenance is implicit.** MIST's "processed_solar_dataset.csv" is, value for value, the 2010–2013 ≥M training file of the 2019 LSTM study. LLMFlare's test set is byte-identical to MIST's. Neither relationship is documented.
  2. **Split design is the dominant source of optimism.** The LSTM Table 2 split (chronological, AR-disjoint) is exemplary. Datasets split at random over overlapping windows give inflated skill: MIST TSS 0.906 falls to 0.197 on independent years. Bundled "test" files sometimes overlap training files: in SolarFlareNet, 84 % of the 24 h test rows also appear in training.
  3. **Missing values are encoded as 0** in the SHARP-based tables, so they cannot be distinguished from legitimate zeros (e.g., flare-history counts). Sentinel fill values (9999.99 etc.) remain in some solar-wind tables.
  4. **Formats are inconsistent.** Timestamps such as `2010-1-1-0`, `1/1/2011 1:34` and `2014-12-28T01:22:09.20Z` are mixed. Some labels are stored as text (`Positive`, `P`, `p`, `padding`, flare classes). Features are selected by column position in several loaders, which already caused one silent column shift (SolarFlareNet).
  5. **Persistence risk.** Several models and inputs are hosted only on `web.njit.edu` (FibrilNet, SDNN), Google Drive (SolarCNN) or `nature.njit.edu` (MagNet, whose package is missing a code module), with no checksum. DeepGeoCME (GitHub release plus a SHA-256 manifest) and the Zenodo DOIs are the best practice in the collection.
  6. **Column-order inconsistency.** RNN-CME ships the same samples twice, with the 18 SHARP columns in different orders. Only one ordering is physically plausible, and the released models were trained on the other. A data dictionary and a column-sanity test would have caught this.
  7. **FlareDB (the CCDC dataset).**
     - It is an *event* database: 151 ≥M5.0 flares within 50° of disk centre, May 2010–May 2025, 82 ARs; HMI/AIA patches in HPC and CEA; 5,285 quick-look movies on Zenodo.
     - It has no non-flare examples, so it supports case studies, flare-evolution modelling and self-supervised learning, but not flare/no-flare classification on its own.
     - The GitHub event list is an earlier subset (103 flares, 57 ARs, to 2024-05), with no header, mixed time formats and 45 rows missing two columns.
     - The Zenodo rights metadata lists both CC-BY-4.0 and "All rights reserved".
- **Funding context.** NSF's *Unlocking Dataset Value for AI-Enabled Scientific Discovery (AI Datasets)*, **NSF 26-512**, is open. The full-proposal deadline is **4 November 2026**, then the first Wednesday of November annually (verified on nsf.gov, page updated 5 October 2026). NASA Heliophysics ran **ROSES B.16 "Heliophysics AI/ML-Ready Data" (HARD)** in 2022–2023. No ROSES-24 awards were made, and the topic now falls under **B.5 RADIANT** (ROSES-25). These are public programmes, **not** grants held by Prof. Wang's group.

2. What is known about CCDC and SolarDB

Item	Status	Source
CCDC page: "CCDC: Community Coordinated Data Center"; one dataset card (FlareDB) with GitHub and Citation (Zenodo) links	✅ opened 2026-10-13 (`papers/citations/page_snapshots_2026-10-13/ccdc.html`, `ccdc_pages_data.json`)	page
SolarDB ("A Cyberinfrastructure for Advancing Space Weather Research") maintained by NJIT's Institute for Space Weather Sciences, partially NSF-supported	🔎 search snippet of `nature.njit.edu/solardb/`	search

Item	Status	Source
SolarDB's three ML-ready datasets: (i) a flare database (M5.0+ flares of cycle 24, SDO images from 12 h before to 6 h after each flare), (ii) a global Hα network (8 observatories), (iii) a BBSO data archive	🔎	Nita et al. 2022, EarthCube white paper, arXiv:2203.09544
FlareDB (the CCDC dataset): 151 significant (≥M5.0) flares from 82 ARs, May 2010–May 2025; AIA AR patches in HPC and CEA projections, aligned with HMI magnetogram patches; 5,285 quick-look movies on Zenodo	✅ paper read; GitHub list profiled (§2.1)	Liu N., Abduallah Y., Kapure T., Li Q., Wang H., Wang J. T. L., *Scientific Data* (2026), doi:10.1038/s41597-026-06607-7; Zenodo doi:10.5281/zenodo.16790538
MagNet data (`/solardb/magnetdata/MagNet/`): 12 MDI and 12 HMI validation cases (Hα, LOS, Bx, By, cor), the authors' predicted Bx/By, a 1.18 GB model; titled "Un-published code"	✅ listed and checked; `layers/attention.py` missing	Jiang et al. 2023, Sol. Phys. 298:87; `results/MagNet/`
SolARED (partner): acoustic power, unsigned flux and continuum-intensity time series for 50 emerging ARs (2010–2023), 81 tiles, FITS download via `sun.njit.edu/sarportal`	🔎	Kasapis et al. 2026, arXiv:2601.13145
Interactive Multi-Instrument Database of Solar Flares (`solarflare.njit.edu`)	🔎 exists	search

The earlier hypothesis that CCDC is a "Community Coordinated *Data* Center", by analogy with CCSC, is confirmed by the page title. Its catalogue currently holds only FlareDB.

2.1 FlareDB: data-science profile

Sources: the *Scientific Data* paper (full text), the Zenodo/DataCite record and the GitHub repository (`scripts/datasets/profile_flaredb.py` → `results/datasets/flaredb_profile.json`).

Question	Answer
Source	SDO/HMI (`hmi.sharp_720s`, vector field, LOS magnetogram, continuum) and SDO/AIA (UV 1600/1700 Å; EUV channels) from JSOC; GOES flare list for event selection
Selection	All flares ≥ M5.0 within 50° of disk centre, May 2010–May 2025: **151 events from 82 ARs** (the paper calls this complete for its criteria)
Time coverage per event	HMI and 12-min AIA: $T_0 - 24$ h → $T_0 + 8$ h; full-cadence AIA (12/24 s): a shorter window around the flare
Processing	SunPy; AR patches in HPC and CEA projections aligned between AIA and HMI; vector fields standardised to Bx, By, Bz; standard JSOC FITS; 13 data types
Distribution	Zenodo: **5,285 quick-look movies** (35 per event) in 35 zip files, plus an event list (PDF) and a data-structure figure. GitHub: download, processing and movie scripts and a CSV event list. The full FITS cubes are described as downloadable from SolarDB
GitHub event list	`Flare_event_list.csv`: **103 rows, no header** (inferred columns: class, NOAA AR, start, peak, end, HARPNUM, two coordinate-like columns). 62 M + 41 X flares (min M5.0), 57 ARs, peaks 2011-02-13 → 2024-05-11, none in 2016 or 2018–2020. **45 rows lack the last two columns**; two time formats (`HH:MM` and `HH:MM:SS.s`). Last commit 2024-12-10, so it is an earlier subset of the 151 events
Multiplicity	19 ARs have more than one event, up to **10 flares in one AR**
Labels	Event-level metadata (GOES class, times). **No negatives** (non-flaring ARs or quiet intervals)
Licence and metadata	Paper: CC-BY 4.0. The DataCite record lists **both** CC-BY-4.0 and "Copyright … All rights reserved", and gives no sizes or formats

How FlareDB supports ML, and its pitfalls.

- *Good fit:* case studies; multimodal representation learning (aligned HMI + AIA cubes); pre-/post-flare evolution modelling (e.g. predicting the AIA response from the HMI evolution); flare-onset timing *given* a flare; benchmarking image-translation tools such as MagNet, SolarCNN and DeepHMI on aligned patches.
- *Not on its own:* flare/no-flare forecasting, because every sample is a large flare (selection bias). Negatives must come from elsewhere (e.g. SHARP series of non-flaring ARs), with matched selection criteria (disk position, data quality).
- *Leakage risk:*
  - Events cluster by AR (up to 10 per AR), and their −24 h/+8 h windows overlap in time for closely spaced flares.
  - Splits must therefore be **grouped by AR and time-blocked**, not random by event.
  - The 50° disk-centre criterion should be applied identically to any negative set.
- *Missing values and cleaning:* the CSV needs a header, a single ISO-8601 time format, and an explicit missing-value marker for the two incomplete columns. The paper documents the processing steps (projections, value ranges), which is good AI-ready practice.

3. Inventory of ML-ready datasets actually inspected (bundled with CCSC tools)

Source of all numbers: `results/datasets/bundled_dataset_profile.json` and the per-tool audits.

Dataset (file)	Origin of raw data	Format	Size	Coverage	Unit of sample	Labels	Split as shipped
LSTM Table 2 (≥C/ ≥M/≥M5.0)	SDO/HMI SHARP (JSOC) + GOES flares (NOAA NCEI)	CSV, 40 features + metadata	train 84,577 / val 26,473 / test 44,689 rows	2010-05 → 2018-05	AR × hour	flare class in next 24 h	**chronological, AR-disjoint** (0 shared HARPs)
LSTM Table 5 (≥C, incl. incomplete samples)	same	CSV (normalised)	127,709 rows (train)	2010-05 → 2014-01	AR × hour	same	as Table 2
SolarFlareNet	SHARP + GOES	CSV, 25 SHARP + metadata	train ≈ 180–222k, test ≈ 63–78k rows per class/window	2010-05 → 2022-09	HARP × 12 min/1 h	N / P / p ("p" skipped)	test (2022) **contained in training** for 22–84 % of rows
SEP	SHARP + GOES + DONKI (flare–CME–SEP links)	CSV	train 7,002 / test 801 rows (24 h, FC_S)	2011 → 2017	HARP × time	SEP yes/no	chronological; **test = 3 ARs (2 positive)**
RNN-CME	SHARP + GOES + DONKI	CSV (normalised)	75,063 rows (24 h train)	2010 → 2019	HARP × 12 min	N / P / padding	test = **2 ARs**
FlareML	SHARP + GOES	CSV, 13 features	845 flares	2010-05 → 2016-12	flare day	B/C/M/X	demo test (40) overlaps training sample (31/40)
MIST	= LSTM Table 2 ≥M train (verified)	CSV + NumPy windows	84,577 rows → 84,553 windows	2010-05 → 2014-01	24 h window	≥M in next 24 h	**random 80/20 over overlapping windows**
CMETNet	ICME list with CME (LASCO-derived) and solar-wind features — exact catalogue sources per the paper, not verified here; LASCO C2 FITS	CSV + 351 FITS	363 ICMEs, 672 MB images	1996 → 2021	CME event	transit time (h)	leave-one-year-out; best of 10 rounds chosen **on test year**
Dst / Kp	NASA OMNIWeb hourly solar wind + indices	CSV	≈ 96–104k rows	2010 → 2022	hour	Dst (nT) / Kp	Kp: clean temporal; Dst: **test window also inside training frame**
SYMHnet	OMNI 1-min/5-min + SYM-H	CSV	315k (1-min) / 72k (5-min) train rows per storm folder	1998 → 2018, 42 storms	minute	SYM-H	by storm (1–25 train, 26–42 test); 1 h and 2 h folders byte-identical
TSInet	SORCE, TCTE (LASP), NRLTSI2, SATIRE-S/M	CSV	6,017 (SORCE) … 50,038 rows	2003 → 2020 (+ models)	day	TSI (W m^{-2})	train SORCE / test other series
SINet	NOAA F10.7, Toyokawa/Nobeyama F30	CSV + preprocessing pickle	—	test 2009 → 2021	day	index value	train data not shipped
SolarUnet / FibrilNet / SDNN	BBSO/GST (NIRIS, Hα), SWAMIS labels, ME inversion	PNG/FITS	3 / 5 test images; 242 train pairs	2017–2018	image	masks / maps	demo only
SolarCNN / DeepHMI / DeepGeoCME	SOHO/MDI + SDO/HMI; HMI + GST/NIRIS; SOHO + SDO	FITS / PyTorch tensors	3 pairs / 1 pair / 33 events	—	image / event	HMI / NIRIS / geoeffective	demo only (SolarCNN's full set is in a restricted-access Zenodo record; the DeepHMI and DeepGeoCME archives hold only the shipped files)
RNN-CME `CME_data_samples`	as RNN-CME	CSV with header	12–60 h: 20,347–60,769 train / 1,312–3,297 test rows (= paper Table 1)	2010 → 2019	HARP × 12 min	N / P	the paper's full split; **column order differs from the training copy**
MagNet validation (SolarDB)	SOHO/MDI, SDO/HMI, BBSO Hα	FITS	12 MDI + 12 HMI cases, ≈ 35 MB	2010–2011	AR image	Bx, By	validation only
FlareDB event list (CCDC)	GOES + JSOC	CSV, no header	103 events (paper: 151)	2011-02 → 2024-05	flare event	GOES class	none

4. Answers to the questions in the brief

Where do the data come from? Mostly NASA/NOAA archives:

- SDO/HMI SHARP parameters from JSOC (Stanford), SDO/AIA images;
- GOES X-ray flare catalogues (NOAA NCEI); flare–CME–SEP associations from NASA DONKI;
- solar wind and geomagnetic indices from NASA OMNIWeb/SPDF; LASCO catalogues;
- TSI from LASP (SORCE, TCTE) and model reconstructions (NRLTSI2, SATIRE);
- ground-based BBSO/GST data from NJIT's own observatory, and the global Hα network.

Formats. CSV for tabular time series (one row per AR-time or time step); FITS for images and magnetograms; PNG for demo masks; NumPy, Pickle and PyTorch tensors for derived windows and models. Model weights are HDF5, TF checkpoints or SavedModel, and `.pth`.

Size. From kilobytes (845-flare table) to several GB (SYMHnet storm folders, ≈ 3.7 GB; SolarFlareNet, ≈ 1.9 GB). The full image datasets (SolarCNN, DeepHMI, FlareDB) are on Zenodo or SolarDB.

Collection, cleaning, integration. Typical pipeline, reconstructed from the code:

1. Download SHARP time series per HARP.
2. Attach NOAA AR numbers (`harp_to_noaa.csv`).
3. Join GOES flare events by AR and time to create look-ahead labels (e.g. "≥M flare within 24 h of t").
4. Derive flare-history features (counts and decayed sums of past flares).
5. Z-normalise.
6. Drop samples whose physical features contain 0 (the convention for "missing").
7. Build fixed-length sequences per AR, zero-padding the history.

For geomagnetic indices: merge OMNI solar-wind parameters with the index at a time offset, then remove sentinel fill values.

Missing values and quality issues (`results/datasets/bundled_dataset_profile.json`):

- **0 is the missing-value code** for SHARP features. In LSTM Table 2, 27 % of physical-feature cells are exactly 0, many of them legitimate zeros in flare-history counts. The `has_zero_record` filters therefore conflate "missing" and "zero", and one of them checks the wrong row (§ LSTM in the main report).
- **Sentinel values** remain in some solar-wind tables: 0.13 % of cells in the Dst training file equal 9999.99-type fill values.
- **Non-standard timestamps:**
  - `YYYY-M-D-H` without zero padding is not lexicographically sortable; NRLTSI2 dates failed to parse in TSInet and fell back to the current date.
  - Time-only uniqueness is not guaranteed: TSInet's NRLTSI2 output repeats the same date.
- **Label encodings vary:** `Positive/Negative`, `P/N/p`, `padding`, raw GOES class strings. Alphabetical label encoding silently flips the positive class for `['N','X']`, which inverted the LSTM ≥M5.0 demo.
- **Positional column selection:** SolarFlareNet's loader assumes another table's column order, so 9 unintended features are used.

Metadata. CSV headers are informative (SHARP keywords), and READMEs describe columns for the LSTM and RNN-CME datasets. However, there are no machine-readable data dictionaries (units, missing codes, valid ranges), no declared split manifest, and no provenance links between derived datasets (MIST ← LSTM Table 2; LLMFlare ← MIST). Several READMEs give software versions but not data versions or SHARP release keywords.

How do the data support ML? They are ready-made supervised learning tables (labels attached, normalised, windowed) with pre-trained models and notebooks, which is excellent for onboarding. What is missing for *reliable* ML is the evaluation scaffolding: recommended splits, baselines, and documented class balance and effective sample size.

Leakage and train/test issues (all verified with scripts):

Dataset	Issue	Effect measured
MIST	random split over 24 h windows overlapping by 23 h; scaler fit on all rows	TSS 0.906 → **0.197** on 2015–2018 (AUC 0.997 → 0.833)
SolarFlareNet	2022 test rows also in training files (22–84 %)	8/9 shipped models have no skill on the test files; C/24 h TSS 0.79 overall vs 0.697 on non-duplicated rows
FlareML demo	31/40 test rows in training sample	demo TSS not out-of-sample; AR-grouped CV lowers macro-TSS from 0.338 to 0.246
LSTM Table 5	threshold chosen on test folds	Table 2 retraining: ≥M5.0 TSS 0.734 (fixed) / 0.825 (validation-chosen) / 0.877 (test-chosen)
Dst	test window inside training frame; labels used in `predict()`	raw model R^2 0.06–0.38 vs reported ≈ 0.94
CMETNet	best of 10 rounds chosen on hold-out year	test-selected MAE 0.4–2.1 h lower than the average round

Dataset	Issue	Effect measured
SEP / RNN-CME	test sets of 3 and 2 ARs	very wide uncertainty on TSS ≈ 0.86–1.00

5. AI-readiness scorecard of the bundled datasets

Criteria are adapted from the ESIP "AI-Ready Data" checklist and from Poduval et al. (2023, *Front. Astron. Space Sci.* 10, doi:10.3389/fspas.2023.1203598). ● = good, ◐ = partial, ○ = missing.

Criterion	LSTM T2	SolarFlareNet	MIST	SEP	Dst/Kp	SYMHnet	CMETNet	Image sets	FlareDB (CCDC)
Persistent identifier (DOI)	● Zenodo	●	● (Zenodo, CCSC page)	●	●	●	●	◐	● (Zenodo + paper)
Data dictionary (units, missing codes)	◐ README	○	○	○	○	◐ ReadMe	○	○	◐ (paper tables; CSV has no header)
Provenance (raw source, versions, code)	◐	◐	○ (undocumented reuse)	◐	◐	◐	◐	◐	● (paper + scripts)
Explicit split manifest, grouped by time/AR	●	○	○	◐	◐	● (by storm)	◐	○	○ (event list only)
Leakage-free as shipped	●	○	○	●	Kp ● / Dst ○	◐ (horizon)	◐	n/a	n/a (no task defined)
Baselines provided (persistence/climatology)	○	○	○	○	○	◐ (GBM)	◐	○	○
Integrity (checksums)	○	○	○	○	○	○	○	DeepGeoCME ●	◐ (Zenodo keeps per-file MD5; not verifiable here)
Licence stated consistently	◐	◐	◐	◐	◐	◐	◐	◐	◐ (CC-BY vs "All rights reserved")
Hosting redundancy	● GitHub	●	●	●	●	●	●	FibrilNet/SDNN/MagNet ○	● (Zenodo + GitHub + SolarDB)

6. NSF/NASA AI/ML-ready data programmes (public information)

- **NSF 26-512 – Unlocking Dataset Value for AI-Enabled Scientific Discovery (AI Datasets).**
  - Supports making *existing* scientific datasets usable for AI: AI-based feature extraction, metadata generation, dataset integration, robust automated-analysis pipelines, augmentation and harmonisation, and assessment of dataset readiness.
  - Encourages the use of national platforms (NSF data platforms, NAIRR).
  - Full proposals are due **4 November 2026**, then the first Wednesday of November annually.
  - Award tiers: 10–20 Impact awards of up to $2 M, 5–10 Flagship awards of up to $5 M, and 10–20 Planning grants of up to $200 k.
  - Proposals must also address dataset security, integrity and community governance.
  - Source: the nsf.gov programme page (published 21 July 2026, updated 5 October 2026), opened 2026-10-13.
- **NASA ROSES-2022 B.16 Heliophysics AI/ML-Ready Data (HARD), NNH22ZDA001N-HARD.**
  - Aimed at new tools and methods that produce AI/ML-ready datasets from existing heliophysics mission data, including cleaning, missing-value handling and outlier removal.
  - The ROSES-23 Step-2 deadline moved to 2 May 2024. For ROSES-24 NASA announced no new awards and returned proposals without review.
  - For ROSES-25 the scope moved to **B.5 RADIANT**.

    - NJIT's research office circulated the 2022 call (research.njit.edu).
- **Community guidance:**
    - ESIP Data Readiness Cluster, "Checklist to Examine AI-readiness for Open Environmental Datasets".
    - Poduval et al. 2023 (AI-ready data in space science and solar physics: problems, mitigation and action plan). It stresses documentation of processing steps, uncertainties, biases and instrument degradation, plus standardised metadata.
    - FAIR principles.
- **Caveat:** these are funding opportunities and community recommendations. Nothing here implies that Prof. Wang's group holds or has applied for any of them.

7. Research questions this suggests (Data Science angle)

1. **Split manifests as a first-class AI-ready artefact.** Can every SolarDB/CCDC dataset ship a time- and AR-grouped split, together with a leakage test suite that runs automatically? MIST, SolarFlareNet and FlareML show that the measured gain is often larger than the model differences being claimed.
2. **Provenance graphs across derived datasets.** Automatically detect value-level reuse between datasets, as was done manually here (MIST = LSTM Table 2). Record it as machine-readable lineage (PROV-O, RO-Crate).
3. **Missing-value semantics for SHARP-derived tables.** Replace "0 = missing" with explicit masks and quality flags, and measure the downstream effect on flare-prediction skill.
4. **External, cross-instrument validation** (the question raised in the applicant's first e-mail). Stratify MagNet/SolarCNN/DeepHMI errors by instrument (MDI, HMI, IVM, GST), observing conditions, field strength and coverage. Use FlareDB's aligned HMI/AIA patches as a testbed.
5. **Making FlareDB a forecasting benchmark.** Add matched negatives (non-flaring or sub-M5 ARs under the same 50° disk-centre and quality criteria) and AR-grouped, time-blocked split manifests. Can AIA-derived precursors in the $T_0 - 24$ h window improve on SHARP-only models (e.g. the 2019 LSTM) once leakage is controlled?
6. **Column-order and schema tests as part of AI-readiness.** The RNN-CME permutation was detectable in seconds from physical correlations (extensive SHARP parameters vs USFLUX). Such checks could ship as a data-validation suite with every CCDC dataset.
7. **Baselines and effective sample size.** Standardise persistence/climatology references and event-level bootstrap confidence intervals, so that results such as "TSS 0.98 on 2 ARs" are reported with honest uncertainty.
8. **Calibration of Bayesian UQ.** Do the Dst/Kp/SYMHnet uncertainty bands achieve their nominal coverage? The current code scales them with ad-hoc constants.

8. References

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